CNC Procedure 2015_007

Jun 3rd

2015

MPCB Replacement in FS0iMate D Series Controller

MPCB replaced to eliminate "Battery alarm occurs to a short period"

A battery to back-up memory consumes in a short period.

When the battery to back-up memory is consumed, "BAT" warning occurs during CNC running. When it's consumed any more, the contents of the memory can't be maintained.
In this situation, there is possibility that system alarm 500 (SRAM data error) occurs, when CNC power is turned off/ on. Because the contents of the memory will be lost.

Condition of the failure to occur: There is possibility that the following main board for Basic unit C (A02B-0321-B530) which is shipped From April, 2014 to March 6, 2015.

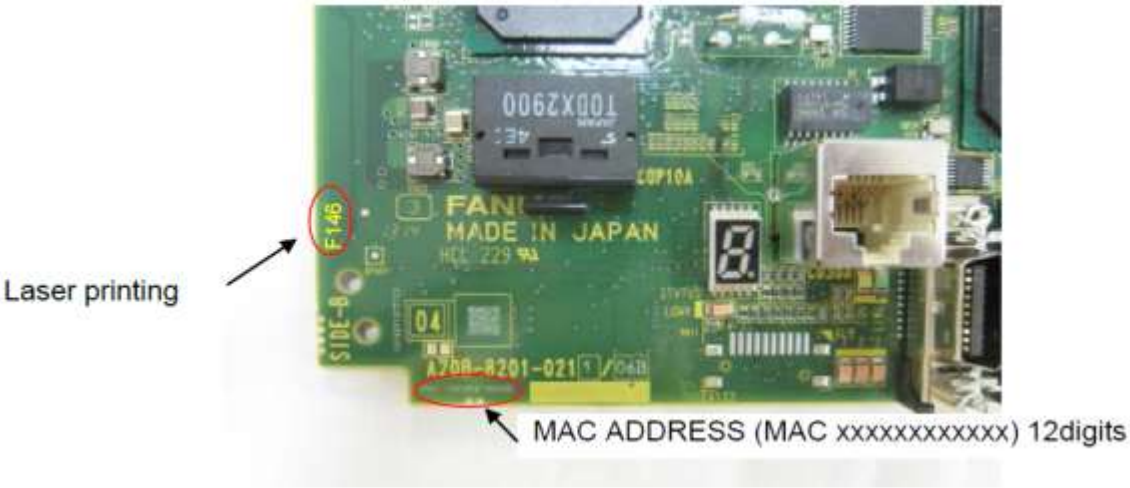
Name	Order specification	PCB specification
Main board C5U	A02B-0322-H135	A20B-8201-0210
Main board C6U	A02B-0322-H136	A20B-8201-0211
Main board C7U	A02B-0322-H137	A20B-8201-0212

Distinction method of target equipment

(1) Confirm a serial number of CNC controller

The target number: E144J0093 - E14ZJ2479, E151H0001～E151H0451, E151J0001～E152J1352 E152H0194, E152H0265, E152H0448, E152H1205, E152H1235, E152H1248 E152H1249, E153H0351, E153H0352, E153H0534, E153J0727

(2) Confirm manufacture year and date by laser print on the main board, when the first alphabet of the serial number of CNC controller is not “E”



3. Cause of the failure:

There is possibility that the Main boards manufactured in period of above target have a part (condenser) added stress because of a problem on the production. Insulation resistance of the part degrades by this. The battery to back-up memory consumes in a short period. Degradation of the part tends to accelerate by the environment of the high humidity.

4. Countermeasures (Modification specification, parts list, revise history, etc.):

When the "BAT" warning which indicates a decline of the battery voltage is displayed in a short period (within about 1 year) after Fanuc shipment and the system alarm 500 (SRAM data error) occurs at power on, check the manufacturing date of the main board. If they're relevant CNC controllers, please exchange the main board.
Measure main boards are supplied separately, so please return an exchanged main board to headquarters.

To presume a failure rate in the field, please send the date and the serial number or the MAC address on the main board which CNC controller was failed to Global service head office monthly until the end of March, 2016.

Tools Required:

1. Star headed screw drive – Long type – 1No.
2. Battery with Ammeter – 1 Set
3. CF Card (Min Size 512 MB) – 1No.
4. 7mm Box spanner with long extension – 1No.
5. Anti-Static rubber glow – 1 Set/customer
6. Multimeter – 1 No.
7. Laptop PCMCI card reader to store the controller Backup (if necessary)

Few parts are shown in snapshot below:

Preparation for MPCB replacement:

Step 1: (NC Data Backup):

(Please do carefully since, we are attending healthy production machine for suspected part replacement).

Please make sure the machine is operational in all respects (If replacement done on good working machine).

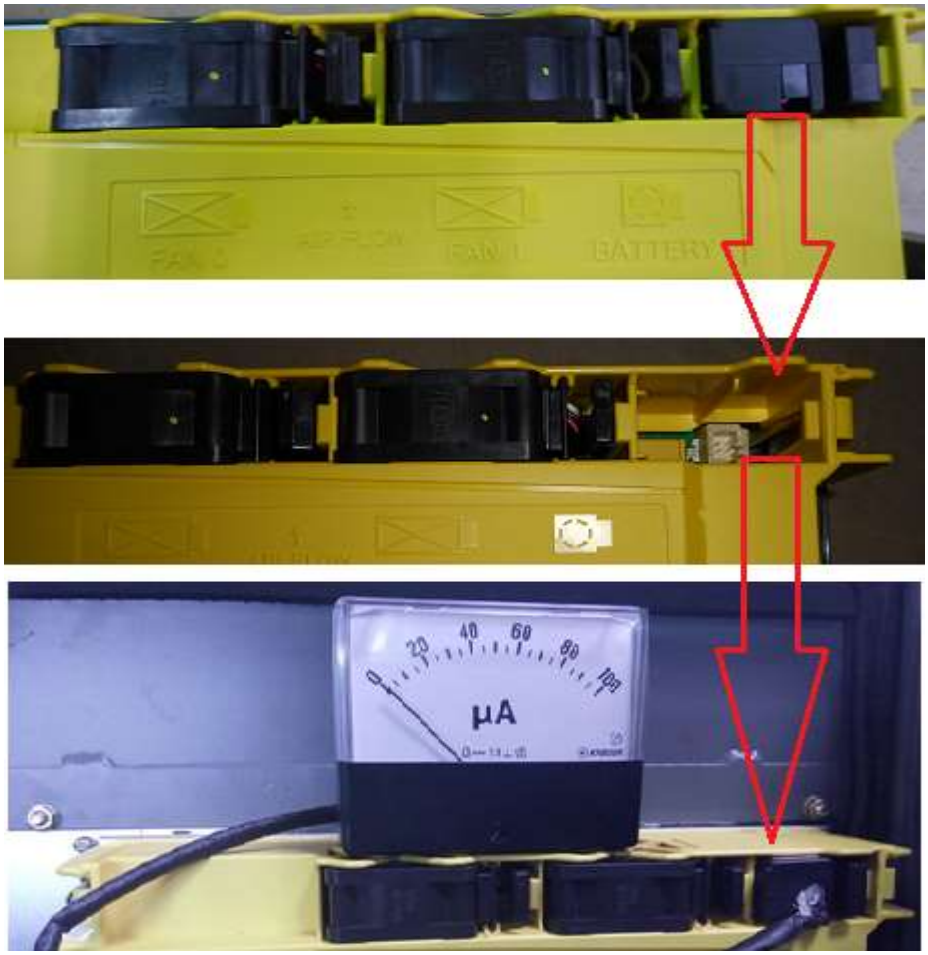
Please collect necessary SRAM/FROM Backup by below procedure with help of CF Card:

1. Switch off the controller if its ON
2. Insert the CF card to PCMCI port on the front panel with correct orientation as shown below:

3. Power On the controller by pressing the left and Right Arrow keys (Softkey below LCD) to get the Boot ROM menus as displayed below::

4. Please format the CF Cards using MEMORY CARD FORMAT menu and operation completes with message MEMORY CARD FORMAT COMPLETED ***(Make sure you are not formatting FROM)***
5. Collect the SRAM data backup using SRAM DATA UTILITY and storing complete with below screen:

6. Exit from the Boot ROM screen and wait till the CNC boot completes and CNC screens are displayed.
7. Check the MPCB condition before processing further by connecting the battery with ammeter setup to the CNC backup battery slot as below:



8. Making judgment on present condition of MPCB based on the Ammeter reading likely:
   - Normally in Power On condition the battery current reads near to **Zero (0) µA** record the readings to the excel sheet.
   - Know switch Off the CNC Controller and carefully check for the ammeter needle position
     - ✓ ~~Ammeter Reading noted is <10µA the PCB is good and may not require replacement~~
     - ✓ **Ammeter Reading noted is >10µA, the PCB needs to be replaced immediately.**

     - ✓ **Record the ammeter reading to the excel data sheet provided.**

~~9. Reconfirm the reading by Switch On the controller and wait till CNC screen appears, check the battery current in the ammeter the value should read near to Zero.~~

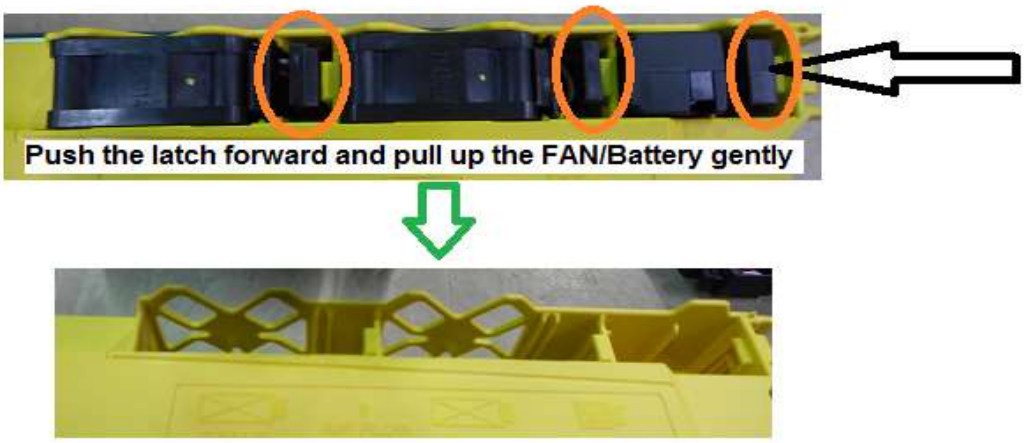
~~10. if the battery current recorded is <10µA, Check the battery voltage Restore the original CNC backup battery unit to the controller with correct orientation~~

~~**11. If ammeter reading continuous to show >10µA, switch Off the controller and progress for MPCB replacement.**~~

12. Steps to replacing MPCB in controller.

- Ensure the controller or machine power is Switched Off with necessary information to machine operator or maintenance person of end-user.
- Please make sure the cable disconnected from controller are clearly marked with permanent markers before disconnecting from the controller for easy re-connecting.
- Use 7mm Box Spanner with long extension to loosen the controller mounting stud screws.
- Please make sure the controller is hold in one hand so the unit is safe from falling away.
- Remove the controller carefully from the fixed location and make sure the necessary packing on the panel is not damaged.
- Take Controller unit to safe work area and palace it on the flat surface as shown below:

- Check for any abnormalities like foreign particle deposition or presence of any liquid media, immediately record your observation and update the end-user accompanying you in this call.
- Clean the controller unit with clean waste or soft brush and remove the foreign particles deposition on the surface. Advise on required corrective action on the operator panel to end-user to avoid any failures in future.
- Remove the controller part by below steps: (Make sure you are wearing Antistatic Rubber Gloves at this point of time):
  STEP 1 – Removing of FANS & BATTERY from the controller



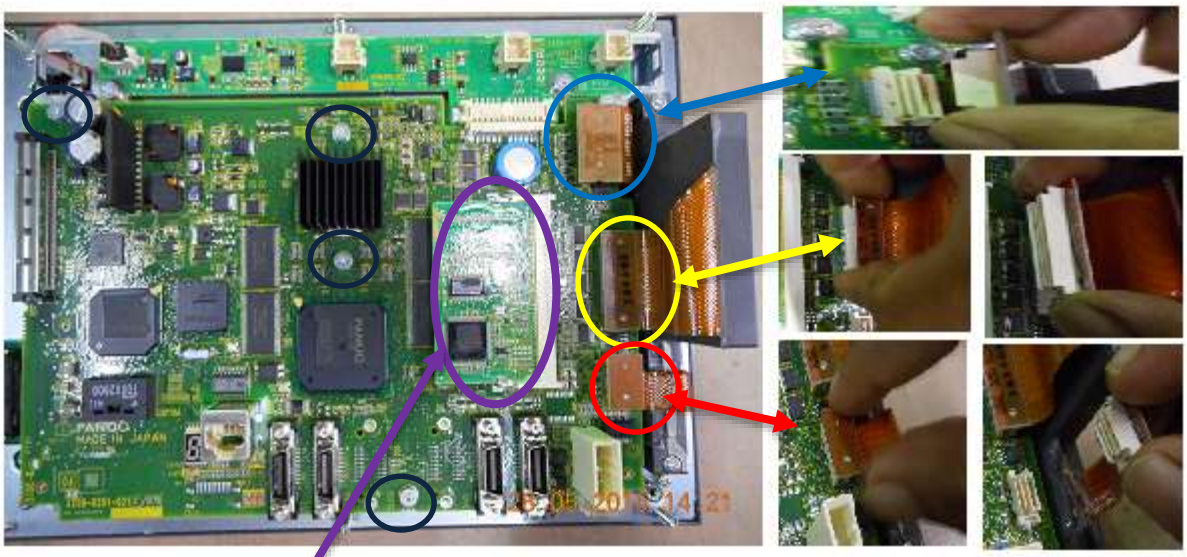
STEP 2 – Remove the MDI connection cable

STEP-3 – Removing Yellow cover

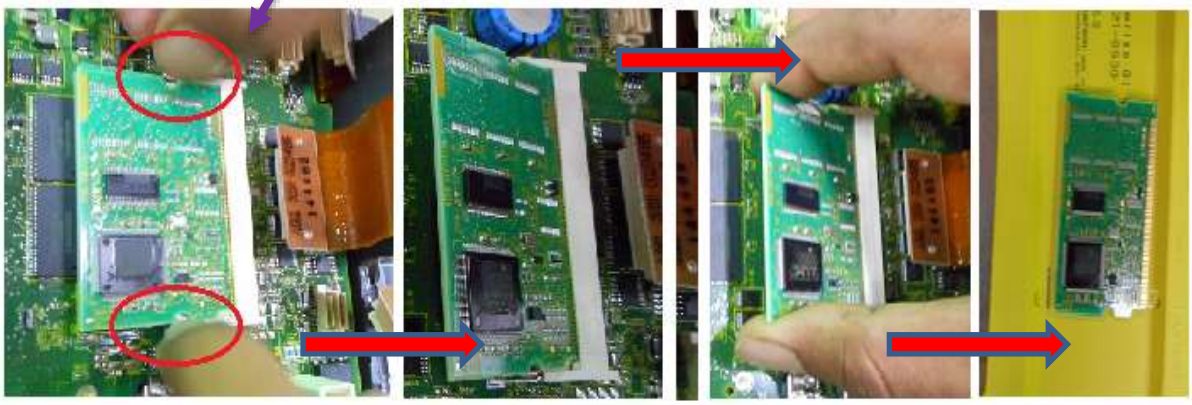
Take care while lifting up the yellow cover for controller, ensure no underneath ribbon cables are safe and place the cover in a safe place near to you.

At this condition the complete CNC hardware is exposed to atmosphere and extra care to be taken to protect from any fall elements nearby your work bench.

STEP-4: Removing of Ribbon Cables used for connecting USB/CF Port/LCD and FROM PCB:



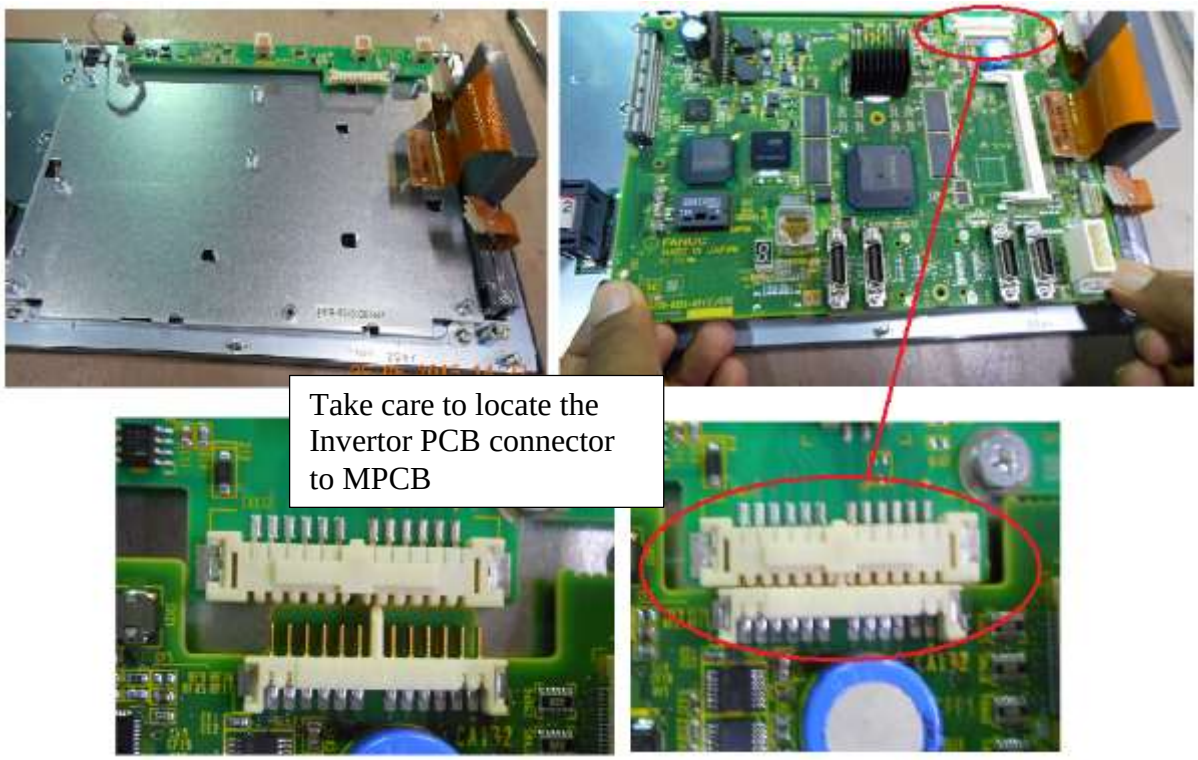
Removing FROM from MPCB



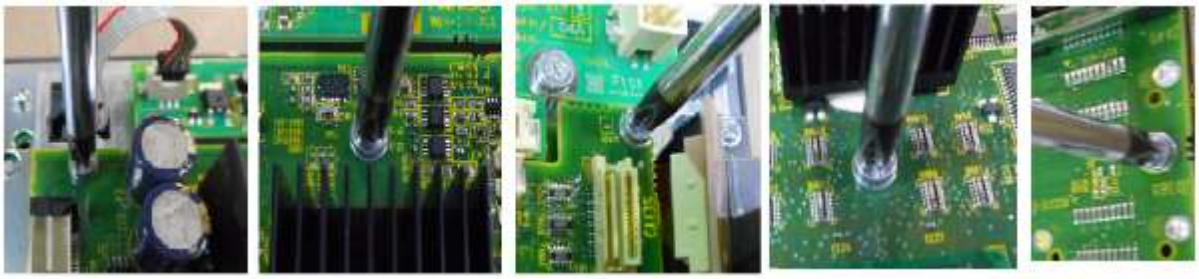
Removing MPCB mounting Screws (5 Nos.):

STEP 5: Remove the MPCB from Controller:

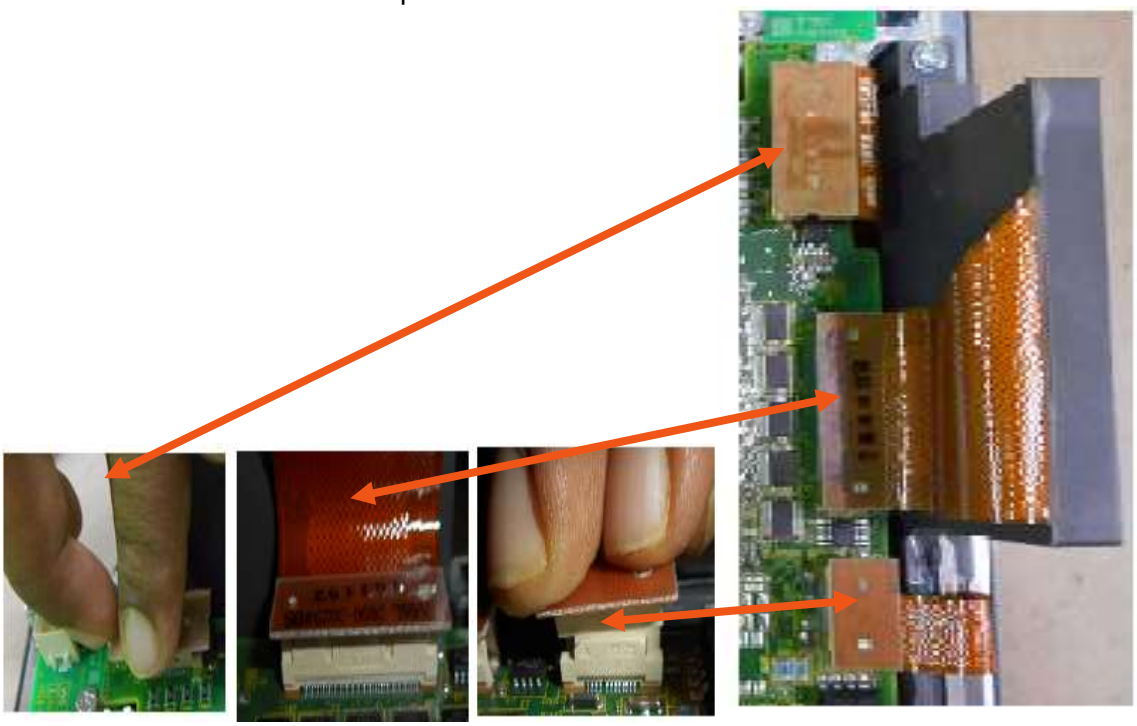
STEP 6: Place New MPCB in the metal case unit



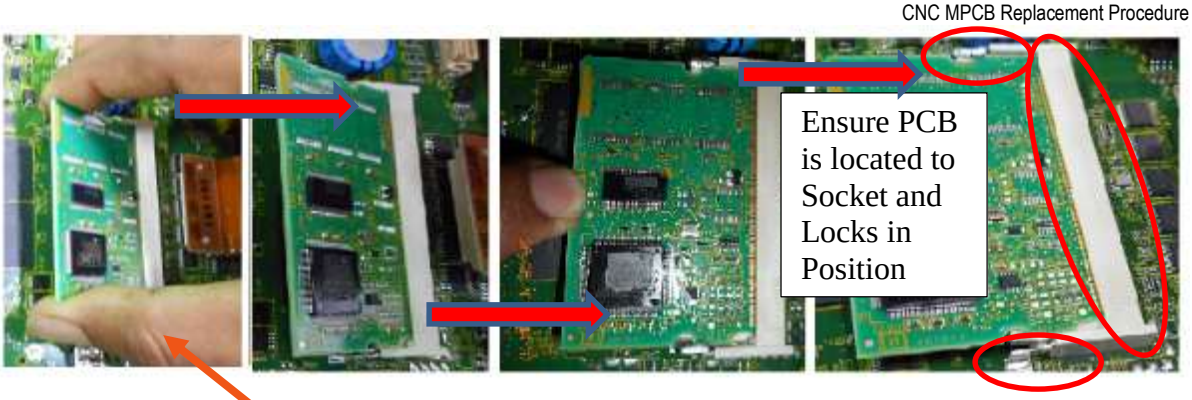
Fasten MPCB mounting screws (5 Nos.) at the below location



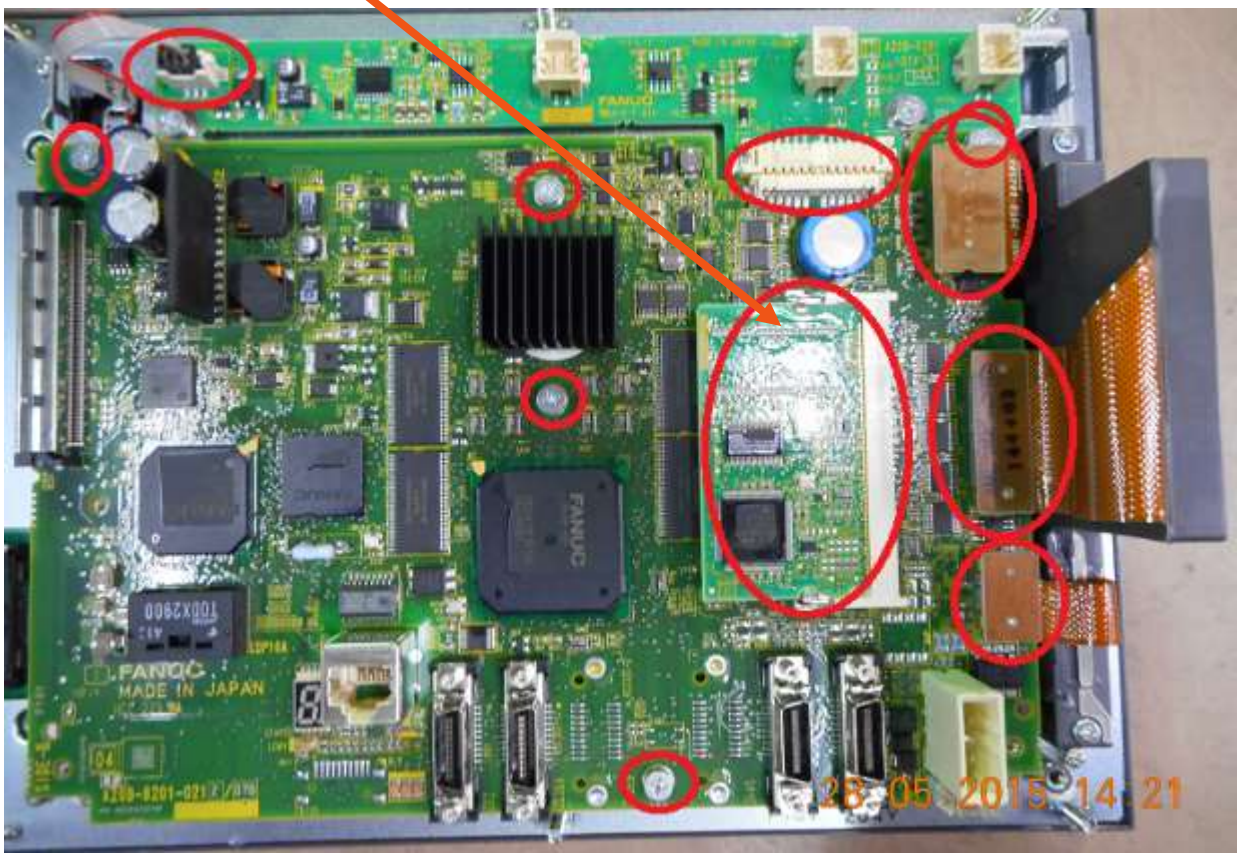
Locate the Ribbon Cables to respective connectors:



Place FROM PCB and Ribbon Cable the respective sockets:

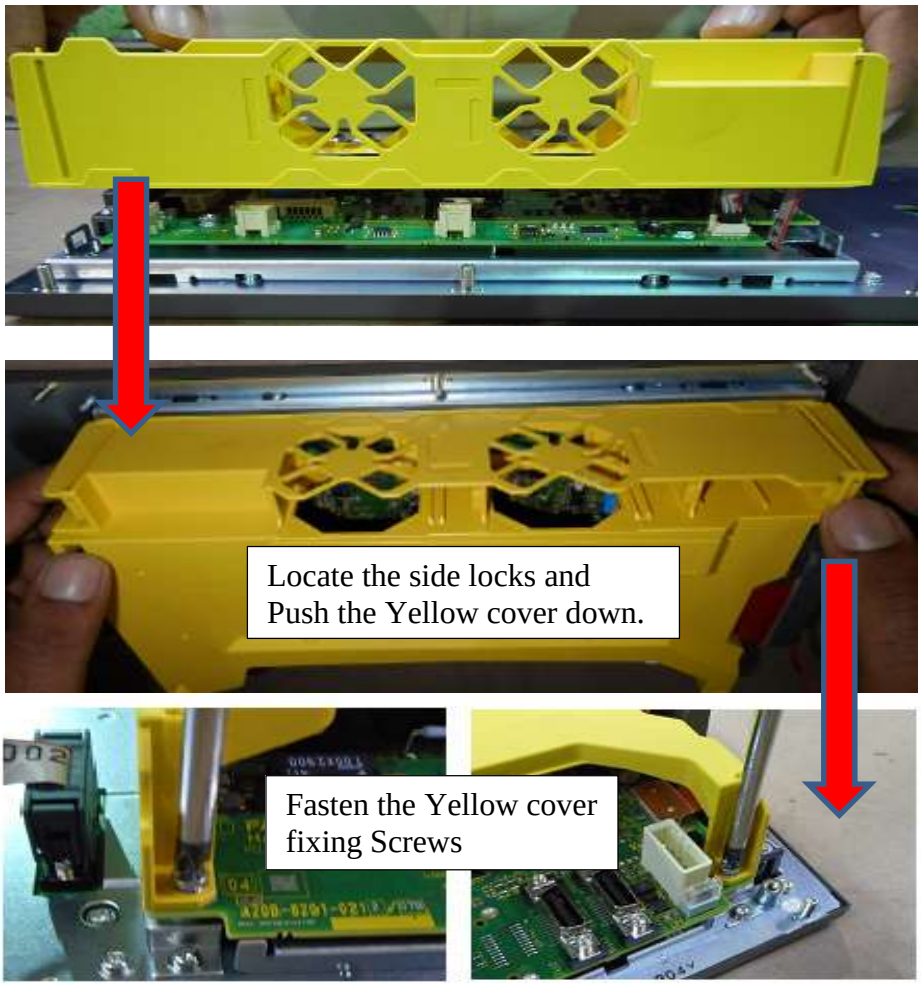


STEP 7: Ensure the circled locations are intact

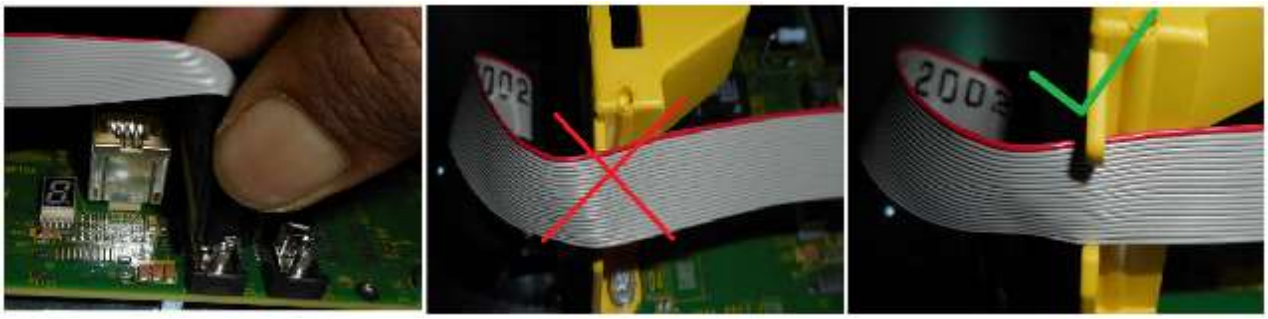


Ensure the circled parts are properly fixing before proceeding further

STEP 8: Mounting Yellow Cover



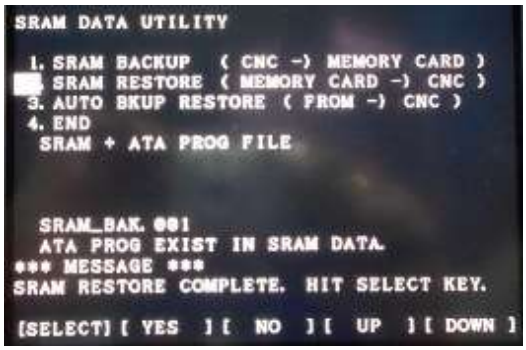
STEP 9: Connecting MDI interface cable



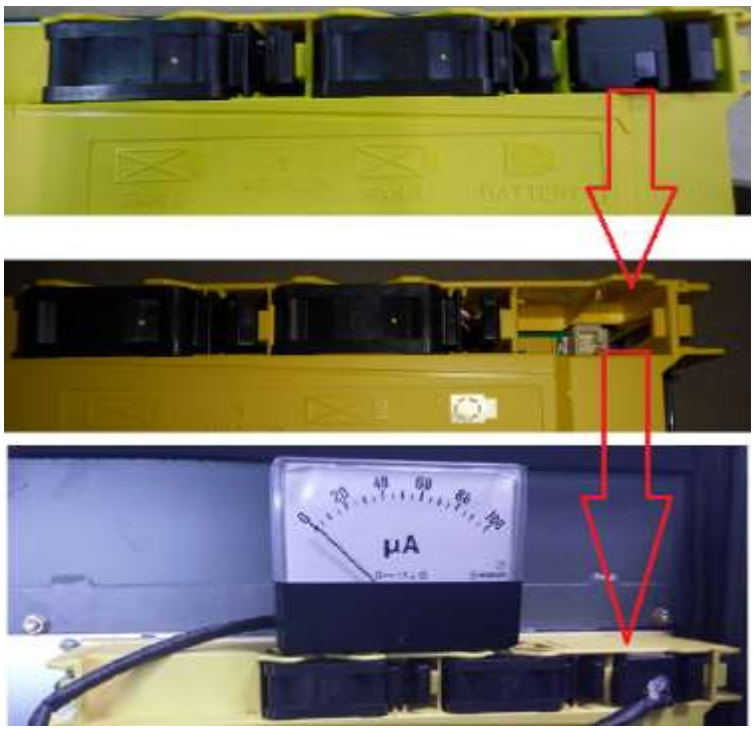
STEP 10: Please check the existing battery voltage and ensure the voltage is not <2.8VDC, If measure low please replace the battery with new part:

STEP 11: Once again ensure all the Fans, batteries and cables are intact

STEP 12: Power On the controller and restore the SRAM data to the controller. Please make sure below message is noticed for successful completion of SRAM data restore.



STEP 13: Check the condition of new replaced MPCB, Allow the controller in ON condition for ~10mins, Connect the battery with ammeter setup to the CNC backup battery slot as below:



STEP 14:

Making judgment on MPCB based on the Ammeter reading likely:

- Normally in Power On condition the battery current reads near to **Zero µA**
- Know switch Off the CNC Controller and carefully check for the ammeter needle position
  - ✓ **Ammeter Reading noted is <10µA the MPCB replaced in good.**
  - ✓ **Ammeter Reading noted is >10µA, the MPCB replaced is also defective.**

- ✓ Record the ammeter reading to the excel data sheet provided.
- ✓ Reconfirm the ammeter reading in Controller Off and On condition.

The summary to judge the MPCB and Battery status is as below:

Check Point	Controller Status	Meter Reading	MPCB	Battery
Battery Current	ON	Near to Zero	OK	OK
		>10µA	REPLACE	REPLACE
	OFF	<10µA	OK	OK
		>10µA	REPLACE	REPLACE
Battery Voltage	Check and replace measured value <2.8VDC			

Please update respective Front Office, in-case the new part replaced is also defective immediately

STEP 15:

Check the machine function in all respect and handover machine to customer for production. Make sure the axis reference points are confirmed OK by machine operator or line in-charge.

Please make proper service report on the activity done and get it signed by end-user authorized person.

End of the activity